

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15A

CHIRALITY, ENANTIOMERS & STEREOCHEMICAL REACTION MECHANISMS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY: CARBONYLS, CARBOXYLIC ACIDS AND CHIRALITY
SUB-TOPIC:	15A: Chirality, Enantiomers & Stereochemical Reaction Mechanisms
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	227 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Chiral Centre Identification****[3 Marks]****WCH14/01/Jan23/Q15**

Lactic acid, 2-hydroxypropanoic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, exhibits optical isomerism.

(a) Define a chiral centre (asymmetric carbon atom). [1]

(b) Identify the chiral carbon atom in lactic acid and draw 3D wedge-dash representations of the two enantiomers. [2]

Q2. 2. Optical Activity & Polarimetry**[4 Marks]****WCH14/01/Oct22/Q16**

Optical isomers rotate the plane of plane-polarised light.

(a) Describe how a polarimeter is used to distinguish between (+) and (−) enantiomers. [2]

(b) Explain why a racemic mixture (racemate) shows zero net optical activity. [2]

Q3. 3. SN1 vs SN2 Mechanism Stereochemical Outcomes**[5 Marks]****WCH14/01/Jun22/Q17**

Nucleophilic substitution of 2-bromobutane with $\text{KOH}(\text{aq})$ can yield either an optically active or inactive product.

(a) Explain why SN_1 hydrolysis of optically active (R)-2-bromobutane yields a racemic mixture. [3]

(b) Explain why $\text{S}_{\text{N}}2$ hydrolysis of (R)-2-bromobutane proceeds with complete inversion of configuration (Walden inversion). [2]

**Q4. 4. Nucleophilic Addition to Unsymmetric Carbonyls: HCN
+ Propanal**

[4 Marks]
WCH14/01/Jan22/Q18

Propanal, $\text{CH}_3\text{CH}_2\text{CHO}$, reacts with HCN in the presence of KCN to form 2-hydroxybutanenitrile.

(a) Explain why 2-hydroxybutanenitrile formed in this reaction is optically inactive. [3]

(b) Identify the nucleophile in this reaction. [1]

Q5. 5. Chirality in Pharmaceutical Drugs: Thalidomide

[4 Marks]
WCH14/01/Oct21/Q16

Thalidomide exists as two optical isomers: (R)-thalidomide cures morning sickness, while (S)-thalidomide is teratogenic.

(a) Explain why it is dangerous to administer racemic thalidomide. [2]

(b) Explain why administering pure (R)-thalidomide still caused birth defects in vivo. [2]

Q6. 6. Number of Optical Isomers: Molecules with Multiple Chiral Centres**[4 Marks]**
WCH14/01/Jun21/Q17

Threonine, 2-amino-3-hydroxybutanoic acid, contains two chiral carbon atoms.

- (a) Calculate the maximum number of stereoisomers given by 2^n . [2]

- (b) Explain what meso compounds are and why they are optically inactive. [2]

Q7. 7. Resolution of Racemic Mixtures**[4 Marks]**
WCH14/01/Jan21/Q17

Racemic mixtures cannot be separated by fractional distillation.

- (a) Explain why enantiomers have identical physical properties (boiling point, solubility, refractive index). [2]

- (b) Describe how a racemic acid can be resolved into pure enantiomers using an optically active base. [2]

Q8. 8. Optical Activity of Hydroxynitriles**[3 Marks]**
WCH14/01/Oct20/Q16

Reaction of ethanal CH_3CHO with HCN forms 2-hydroxypropanenitrile.

- (a) Draw the structure of 2-hydroxypropanenitrile and mark the chiral carbon with an asterisk (*). [2]

- (b) State the optical activity of the product mixture. [1]

Q9. 9. Optical Activity of Substituted Cycloalkanes**[4 Marks]**
WCH14/01/Jun20/Q17

trans-1,2-dimethylcyclopropane is chiral, whereas cis-1,2-dimethylcyclopropane is achiral.

(a) Explain why cis-1,2-dimethylcyclopropane is achiral (meso). [2]

(b) Explain why trans-1,2-dimethylcyclopropane exists as a pair of enantiomers. [2]

Q10. 10. Optical Inactivity of Symmetrical Carbonyl Addition Products**[3 Marks]**
WCH14/01/Jan20/Q17

Propanone, CH_3COCH_3 , reacts with HCN to form 2-hydroxy-2-methylpropanenitrile.

(a) Explain why the product of this reaction does NOT exhibit optical isomerism. [3]

Q11. 11. Chirality of Amino Acids**[4 Marks]**
WCH14/01/Oct19/Q17

All naturally occurring alpha-amino acids except glycine are optically active.

(a) Explain why glycine (2-aminoethanoic acid) is optically inactive. [2]

(b) State which enantiomeric configuration (L- or D-) is present in natural proteins. [2]

Q12. 12. Stereospecific Enzyme Active Sites**[4 Marks]**
WCH14/01/Jun19/Q18

Enzymes usually catalyze reactions of only one specific enantiomer.

- (a) Explain in terms of 3-point active site binding why enzymes are stereospecific. [4]

Q13. 13. Optical Activity of Halogenoalkane Elimination Products**[4 Marks]**
WCH14/01/Jan19/Q19

Reaction of 2-bromobutane with ethanolic KOH forms E/Z alkenes.

- (a) Explain why the alkene products (but-1-ene, E-but-2-ene, Z-but-2-ene) are all optically inactive. [4]

Q14. 14. Chirality of Tartaric Acid (2,3-dihydroxybutanedioic acid)**[5 Marks]**
WCH14/01/sample/Q14

Tartaric acid $\text{HOOC}-\text{CH}(\text{OH})-\text{CH}(\text{OH})-\text{COOH}$ has 2 chiral centres.

- (a) Draw the structures of (2R,3R)-tartaric acid, (2S,3S)-tartaric acid, and meso-tartaric acid. [3]

(b) Explain why meso-tartaric acid is optically inactive. [2]

Q15. 15. SN1 Substitution of Chiral Secondary Halogenoalkanes

[5 Marks]

WCH14/01/sample/Q15

Optically pure (S)-1-phenylethanol is converted to 1-chloro-1-phenylethane with HCl.

(a) If the reaction proceeds via SN1, predict the optical purity of the product. [3]

(b) Explain why partial retention/inversion can occur if solvent cage blocks one face. [2]

Q16. 16. Optical Activity of Carboxylic Acid Chlorination

[4 Marks]

WCH14/01/sample/Q16

Reaction of (R)-2-chloropropanoic acid with PCl5 forms 2-chloropropanoyl chloride.

(a) Does this reaction affect the chiral centre? [2]

(b) State whether the product is optically active. [2]

Q17. 17. Polarimeter Specific Rotation Formula [α]**[4 Marks]****WCH14/01/Sample/Q17**

Specific rotation [α] = $\alpha / (c \times l)$ where c = conc in g cm⁻³, l = path length in dm.

- (a) A solution of 2.00 g D-glucose in 10.0 cm³ water in a 1.00 dm cell rotates light by +13.3°. [3]

- (b) Calculate specific rotation [α]. [1]

Q18. 18. Asymmetric Synthesis via Chiral Catalysts**[4 Marks]****WCH14/01/Sample/Q18**

Modern pharmaceutical synthesis uses chiral catalysts to produce single enantiomers.

- (a) Explain the advantage of asymmetric synthesis over resolving racemic mixtures. [2]

- (b) State how a chiral transition metal catalyst forces reaction down one enantiomeric pathway. [2]

Q19. 19. Chirality of Allenes (1,3-substituted propadienes)**[4 Marks]****WCH14/01/Sample/Q19**

Penta-2,3-diene CH₃-CH=C=CH-CH₃ exhibits optical isomerism despite lacking a tetrahedral chiral carbon atom.

- (a) Explain why substituted allenes possess axial chirality. [4]

Q20. 20. Optical Activity of Reduction of Unsymmetrical Ketones**[4 Marks]****WCH14/01/Sample/Q20**

Reduction of butanone $\text{CH}_3\text{COCH}_2\text{CH}_3$ with NaBH_4 produces sec-butanol (butan-2-ol).

- (a) Explain why the product butan-2-ol is optically inactive (racemic). [4]

Q21. 21. Chirality in Terpenes: Menthol and Carvone**[4 Marks]****WCH14/01/Sample/Q21**

(R)-carvone smells of spearmint, whereas (S)-carvone smells of caraway seeds.

- (a) Explain why two enantiomers can produce completely different olfactory sensations. [4]

Q22. 22. Chirality of Substituted Biphenyls (Atropisomerism)**[4 Marks]****WCH14/01/Sample/Q22**

2,2'-dinitro-6,6'-dimethylbiphenyl exhibits optical activity.

- (a) Explain why restricted rotation around C-C single bond creates atropisomers. [4]

Q23. 23. Cahn-Ingold-Prelog (CIP) Priority Rules for R/S Assignment**[5 Marks]****WCH14/01/Sample/Q23**

Assign priority (1 = highest, 4 = lowest) to -OH, -CHO, -CH₂OH, -H according to CIP rules.

(a) Rank the four groups in order of priority with atomic number justification. [4]

(b) State how R or S configuration is determined when viewing with group 4 pointing away. [1]

Q24. 24. Optical Activity of Cyanohydrin Hydrolysis**[4 Marks]****WCH14/01/Sample/Q24**

2-hydroxybutanenitrile is hydrolysed by dilute HCl to 2-hydroxybutanoic acid.

(a) If the starting cyanohydrin is racemic, state the optical activity of 2-hydroxybutanoic acid. [2]

(b) Write an equation for nitrile acid hydrolysis. [2]

Q25. 25. Diastereomers vs Enantiomers Comparison**[4 Marks]****WCH14/01/Sample/Q25**

Compare enantiomers and diastereomers.

(a) Define diastereomers and state two ways their physical properties differ. [4]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Racemization Mechanism via Enolization****[5 Marks]**
WCH14/01/Hard/Q26

Optically active (R)-2-phenylpropanal undergoes rapid loss of optical activity in acidic or basic solution.

- (a) Explain the mechanism of racemization via planar enol / enolate intermediate. [5]

Q27. 27. A* Challenge: Complete Multi-Step Synthesis of Pure Enantiomer**[5 Marks]**
WCH14/01/Hard/Q27

Synthesize pure (S)-2-hydroxypropanoic acid from ethanal using an enzyme-catalysed step.

- (a) Step 1: Ethanal + HCN in presence of oxynitrilase enzyme $\rightarrow$ (S)-2-hydroxypropanenitrile. [3]

- (b) Step 2: Hydrolysis of nitrile with dilute HCl. [2]

Q28. 28. A* Challenge: Chiral HPLC Separation using Cyclodextrin Stationary Phase**[5 Marks]**
WCH14/01/Hard/Q28

Enantiomers can be separated using chiral High Performance Liquid Chromatography (HPLC).

- (a) Explain how a chiral stationary phase (e.g. beta-cyclodextrin) separates (+) and (-) enantiomers. [5]

Q29. 29. A* Challenge: Stereochemistry of Electrophilic Addition to Alkenes**[5 Marks]**
WCH14/01/Hard/Q29

Addition of Br₂ to trans-but-2-ene yields meso-2,3-dibromobutane, whereas addition to cis-but-2-ene yields a racemic mixture.

- (a) Explain the anti-addition mechanism via cyclic bromonium ion intermediate. [5]

Q30. 30. A* Challenge: Polarimetry Enantiomeric Excess (ee) Calculation**[5 Marks]**
WCH14/01/Hard/Q30

A mixture of (R)- and (S)-2-chlorobutane has an observed rotation of +9.2°. Pure (R)-isomer has $[\alpha] = +23.0^\circ$.

- (a) Calculate enantiomeric excess % ee = (observed rotation / pure rotation) x 100%. [2]

- (b) Calculate percentage of (R)-isomer and (S)-isomer in the mixture. [3]

Q31. 31. A* Challenge: Stereochemistry of Epoxidation and Ring Opening**[5 Marks]**
WCH14/01/Hard/Q31

Epoxidation of cyclohexene with mCPBA forms cyclohexene oxide. Acid-catalysed ring opening with H₂O yields trans-cyclohexane-1,2-diol.

(a) Explain why trans-diol is formed exclusively rather than cis-diol. [5]

Q32. 32. A* Challenge: Kinetic Resolution of Enantiomers**[5 Marks]**
WCH14/01/Hard/Q32

Lipase enzyme selectively hydrolyses (R)-ester 100x faster than (S)-ester in a racemic mixture.

(a) Explain how stopping the reaction at 50% conversion yields pure (S)-ester and pure (R)-acid. [5]

Q33. 33. A* Challenge: Optical Activity of Chiral Sulfoxides and Amines**[5 Marks]**
WCH14/01/Hard/Q33

Sulfoxides $R-SO-R'$ are optically active, whereas simple tertiary amines $R_1R_2R_3N$ invert rapidly at room temp.

(a) Explain why sulfoxides have a stable pyramidal chiral centre (lone pair = 4th group). [3]

(b) Explain why tertiary amines undergo rapid pyramidal inversion (umbrella flip). [2]

Q34. 34. A* Challenge: Chiral Pool Synthesis from Natural L-Amino Acids

[5 Marks]
WCH14/01/Hard/Q34

L-proline is used as an organocatalyst in asymmetric aldol reactions.

(a) Explain the concept of "chiral pool" in organic synthesis. [2]

(b) Explain how L-proline forms an enamine intermediate to direct aldol addition to one face. [3]

Q35. 35. A* Challenge: Stereochemical Proof of S_N2 Inversion using Radioactive Iodide

[5 Marks]
WCH14/01/Hard/Q35

Hughes and Ingold reacted optically active (+)-2-iodooctane with radioactive iodide $^{*}I^{-}$.

(a) Show that the rate of loss of optical activity is EXACTLY TWICE the rate of radioactive $^{*}I$ exchange. [5]

Q36. 36. A* Challenge: Optical Activity of Condensation Polyesters with Chiral Monomers

[5 Marks]
WCH14/01/Hard/Q36

Poly(L-lactic acid) (PLLA) is a biodegradable chiral polymer used in 3D printing and medical sutures.

(a) Explain why PLLA prepared from pure L-lactic acid is crystalline and optically active. [3]

(b) Compare with poly(D,L-lactic acid) (PDLLA) prepared from racemic lactic acid. [2]

Q37. 37. A* Challenge: Optical Activity of Halogenation of Chiral Alkanes

[5 Marks]
WCH14/01/Hard/Q37

Free radical chlorination of (S)-2-chlorobutane yields 2,2-dichlorobutane, 1,2-dichlorobutane, and 2,3-dichlorobutane.

(a) Identify which product is optically inactive due to loss of chiral centre. [2]

(b) Explain why free radical substitution at C2 proceeds via a planar alkyl radical intermediate. [3]

Q38. 38. A* Challenge: Circular Dichroism (CD) Spectroscopy of Protein Secondary Structures**[5 Marks]**
WCH14/01/Hard/Q38

Circular Dichroism measures differential absorption of left- and right-circularly polarised light.

- (a) Explain how CD spectroscopy distinguishes alpha-helices, beta-sheets, and random coils in proteins. [5]

Q39. 39. A* Challenge: Optical Activity of Grignard Reactions with Chiral Aldehydes**[5 Marks]**
WCH14/01/Hard/Q39

Reaction of CH_3MgBr with chiral (R)-2-phenylpropanal leads to diastereomeric ratio (Felkin-Anh model).

- (a) Explain why addition to a chiral aldehyde produces unequal amounts of diastereomers (diastereoselectivity). [5]

Q40. 40. A* Challenge: Optical Activity of Sugar Mutarotation (D-Glucose)**[5 Marks]**
WCH14/01/Hard/Q40

Dissolving pure alpha-D-glucose ($[\alpha] = +112^\circ$) in water results in specific rotation changing gradually to $+52.7^\circ$ (mutarotation).

(a) Explain the equilibrium mechanism of mutarotation between alpha-D-glucose, open-chain glucose, and beta-D-glucose ($[\alpha] = +18.7^\circ$). [5]

Q41. 41. A* Challenge: Optical Activity of Hydroboration–Oxidation vs Oxymercuration of Alkenes

[5 Marks]
WCH14/01/Hard/Q41

Hydroboration-oxidation of 1-methylcyclopentene yields trans-2-methylcyclopentanol.

(a) Explain the syn-addition mechanism of BH_3 across $\text{C}=\text{C}$ double bond. [3]

(b) State whether the product mixture is optically active. [2]

Q42. 42. A* Challenge: Stereochemistry of Beckmann Rearrangement of Chiral Oximes

[5 Marks]
WCH14/01/Hard/Q42

Beckmann rearrangement of chiral ketoximes proceeds with complete retention of configuration at migrating group.

(a) Explain why migration of chiral R group to nitrogen occurs with 100% retention. [5]

Q43. 43. A* Challenge: Chiral Auxiliary Directed Asymmetric Synthesis (Evans Oxazolidinones)

[5 Marks]
WCH14/01/Hard/Q43

Evans chiral oxazolidinones are covalently attached to acyl groups to direct asymmetric alkylation.

(a) Explain the 4-step chiral auxiliary protocol: Attach → Direct → Separate → Cleave. [5]

Q44. 44. A* Challenge: Optical Activity of Oxymercuration–Demercuration of Chiral Alkenes

[5 Marks]
WCH14/01/Hard/Q44

Reaction of 3-methylbut-1-ene with $\text{Hg}(\text{OAc})_2/\text{H}_2\text{O}$ followed by NaBH_4 yields 3-methylbutan-2-ol without carbocation rearrangement.

(a) Explain why mercurinium ion intermediate prevents carbocation hydride shifts. [3]

(b) State the optical activity of the resulting 3-methylbutan-2-ol. [2]

Q45. 45. A* Challenge: Optical Activity of Ozonolysis of Chiral Terpenes**[5 Marks]**
WCH14/01/Hard/Q45

Ozonolysis of (R)-limonene followed by Zn/H₂O reduction yields a chiral dicarbonyl compound.

- (a) Explain why ozonolysis cleaves C=C double bonds without altering chiral centres elsewhere in the molecule. [3]

- (b) Predict whether the ozonolysis product of (R)-limonene is optically active. [2]

Q46. 46. A* Challenge: Optical Activity of Wittig Reaction with Chiral Ketones**[5 Marks]**
WCH14/01/Hard/Q46

Wittig reaction of (R)-3-methylcyclohexanone with Ph₃P=CH₂ yields an optically active alkene.

- (a) Write an equation for the Wittig reaction. [2]

- (b) Explain why the alkene product retains optical activity. [3]

Q47. 47. A* Challenge: Stereochemistry of Sharpless Asymmetric Epoxidation**[5 Marks]**
WCH14/01/Hard/Q47

Sharpless epoxidation converts allylic alcohols into chiral epoxides using $\text{Ti}(\text{OiPr})_4$, tBuOOH , and (+)- or (-)-diethyl tartrate (DET).

(a) Explain how choice of (+)-DET vs (-)-DET delivers oxygen to top vs bottom face of allylic alcohol. [5]

Q48. 48. A* Challenge: Resolution of Racemic Base via Chiral Tartaric Acid

[5 Marks]

WCH14/01/Hard/Q48

Racemic 1-phenylethylamine is resolved using (+)-tartaric acid.

(a) Explain the chemical difference between (+)-amine.(+)-tartrate salt and (-)-amine.(+)-tartrate salt. [3]

(b) Explain how pure (-)-1-phenylethylamine is recovered after crystallization. [2]

Q49. 49. A* Challenge: Optical Activity of Electrocyclic Reactions (Woodward-Hoffmann Rules)

[5 Marks]

WCH14/01/Hard/Q49

Thermal ring-opening of trans-3,4-dimethylcyclobutene occurs via conrotatory motion to yield (2E,4E)-hexa-2,4-diene.

(a) Explain conrotatory vs disrotatory orbital symmetry rules. [5]

Q50. 50. A* Challenge: Complete Chirality & Stereochemistry
Master Synthesis

[6 Marks]
WCH14/01/Hard/Q50

An unknown organic liquid X (C_4H_8O) does not react with Tollens' reagent but forms a yellow precipitate with 2,4-DNPH.

X is reduced by $NaBH_4$ to compound Y ($C_4H_{10}O$). Y is optically active.

(a) Deduce the structure of X and Y with full reasoning. [4]

(b) Explain why Y produced by $NaBH_4$ reduction of X in the lab is optically INACTIVE, whereas Y produced by yeast fermentation of X is optically ACTIVE. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Chiral Centre Identification 1. (a) A carbon atom bonded to four different groups or atoms (1). 1. (b) C2 is chiral (1). 3D tetrahedral structures drawn as non-superimposable mirror images showing -H, -OH, -CH ₃ , and -COOH (1).	[3]
Q2	Q2: 2. Optical Activity & Polarimetry 2. (a) Pass plane-polarised light through the sample solution in polarimeter (1). Enantiomers rotate light by equal angles in opposite directions (+ dextrorotatory, - laevorotatory) (1). 2. (b) Racemic mixture contains equal (50:50) amounts of both enantiomers (1). Rotation by (+) enantiomer is exactly cancelled by (-) enantiomer (1).	[4]
Q3	Q3: 3. S _N 1 vs S _N 2 Mechanism Stereochemical Outcomes 3. (a) S _N 1 mechanism proceeds via a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ can attack the planar carbocation with equal 50:50 probability from top or bottom face (1), forming equal amounts of (+) and (-) enantiomers (racemate) (1). 3. (b) S _N 2 involves concerted backside attack by OH ⁻ on C-Br bond (1), causing complete inversion of configuration (Walden inversion) (1).	[5]
Q4	Q4: 4. Nucleophilic Addition to Unsymmetric Carbonyls: HCN + Propanal 4. (a) Carbonyl C=O carbon is trigonal planar (1). CN ⁻ nucleophile attacks the planar C=O group with equal probability from above or below the plane (1), producing a 50:50 racemic mixture of enantiomers (1). 4. (b) Cyanide ion, :CN ⁻ (1).	[4]
Q5	Q5: 5. Chirality in Pharmaceutical Drugs: Thalidomide 5. (a) (S)-enantiomer causes severe foetal limb deformities (teratogenic) (2). 5. (b) Thalidomide racemizes in vivo (in human blood pH), converting (R)-enantiomer into teratogenic (S)-enantiomer (2).	[4]
Q6	Q6: 6. Number of Optical Isomers: Molecules with Multiple Chiral Centres 6. (a) $n = 2$ chiral centres $\Rightarrow 2^2 = 4$ stereoisomers (2 pairs of enantiomers) (2). 6. (b) Meso compound has internal plane of symmetry (1); top half rotation cancels bottom half, making it optically inactive (1).	[4]
Q7	Q7: 7. Resolution of Racemic Mixtures 7. (a) Intermolecular forces and bonding distances are identical in both enantiomers (2). 7. (b) React racemic acid (d,l-HA) with single enantiomer base (d-B*) to form diastereomeric salts (d-HA.d-B* and l-HA.d-B*) (1). Diastereomers have different solubilities and can be separated by fractional crystallization (1).	[4]
Q8	Q8: 8. Optical Activity of Hydroxynitriles 8. (a) CH ₃ -C*(H)(OH)-CN (1). Asterisk on C2 (1). 8. (b) Optically inactive (racemic mixture) (1).	[3]
Q9	Q9: 9. Optical Activity of Substituted Cycloalkanes 9. (a) Cis isomer has a plane of symmetry passing through C3, making it superimposable on its mirror image (meso) (2). 9. (b) Trans isomer lacks a plane of symmetry, so its mirror image is non-superimposable (2).	[4]
Q10	Q10: 10. Optical Inactivity of Symmetrical Carbonyl Addition Products 10. (a) Product is (CH ₃) ₂ C(OH)CN (1). Central carbon atom is bonded to TWO identical methyl (-CH ₃) groups (1). It lacks a chiral centre (not asymmetric) (1).	[3]
Q11	Q11: 11. Chirality of Amino Acids 11. (a) Alpha-carbon in glycine H ₂ N-CH ₂ -COOH is bonded to TWO hydrogen atoms (not 4 different groups) (2). 11. (b) L-amino acids (1) (S-configuration at alpha carbon) (1).	[4]

Q12	Q12: 12. Stereospecific Enzyme Active Sites 12. (a) Enzyme active site has 3D binding pockets matching specific functional groups of one enantiomer (2). Only the matching (+ or -) enantiomer can dock into all 3 sites simultaneously (1). The mirror image enantiomer cannot bind properly (1).	[4]
Q13	Q13: 13. Optical Activity of Halogenoalkane Elimination Products 13. (a) Elimination forms C=C double bonds (1). Alkenes have sp ² planar carbon atoms (1). No carbon atom in but-1-ene or but-2-ene is bonded to 4 different groups (no chiral centre) (2).	[4]
Q14	Q14: 14. Chirality of Tartaric Acid (2,3-dihydroxybutanedioic acid) 14. (a) (2R,3R) and (2S,3S) are enantiomers (2). Meso isomer drawn with internal mirror plane (1). 14. (b) Meso isomer has an internal plane of symmetry; top C rotation (+α) cancels bottom C rotation (-α) (2).	[5]
Q15	Q15: 15. S _N 1 Substitution of Chiral Secondary Halogenoalkanes 15. (a) S _N 1 forms planar carbocation intermediate (1). Attack from either face gives 50:50 mixture => 0% optical purity (racemate) (2). 15. (b) Leaving group Cl ⁻ shields one face of carbocation temporarily, causing slight excess of inversion product (2).	[5]
Q16	Q16: 16. Optical Activity of Carboxylic Acid Chlorination 16. (a) No, reaction occurs at carboxyl C=O group (-COOH → -COCl); bonds to chiral C2 are untouched (2). 16. (b) Product retains configuration at C2 and remains 100% optically active (2).	[4]
Q17	Q17: 17. Polarimeter Specific Rotation Formula [α] 17. (a) $c = 2.00 / 10.0 = 0.200 \text{ g cm}^{-3}$ (1). $l = 1.00 \text{ dm}$ (1). $\alpha = +13.3^\circ$. 17. (b) $[\alpha] = +13.3 / (0.200 \times 1.00) = +66.5^\circ \text{ cm}^3 \text{ g}^{-1} \text{ dm}^{-1}$ (2).	[4]
Q18	Q18: 18. Asymmetric Synthesis via Chiral Catalysts 18. (a) Avoids wasting 50% of product as unwanted enantiomer (1); eliminates costly separation/resolution steps (1). 18. (b) Chiral ligands on catalyst create an asymmetric steric environment, blocking approach to one face of substrate (2).	[4]
Q19	Q19: 19. Chirality of Allenes (1,3-substituted propadienes) 19. (a) Central carbon is sp hybridized with two perpendicular pi bonds (2). End C=CH-CH ₃ groups lie in perpendicular planes (1). Lacks a plane of symmetry, forming non-superimposable mirror images (1).	[4]
Q20	Q20: 20. Optical Activity of Reduction of Unsymmetrical Ketones 20. (a) Butanone C=O carbon is planar (1). Hydride ion :H ⁻ from NaBH ₄ attacks planar C=O with equal 50:50 probability from top or bottom face (2). Forms equal amounts of (R)-butan-2-ol and (S)-butan-2-ol (racemate) (1).	[4]
Q21	Q21: 21. Chirality in Terpenes: Menthol and Carvone 21. (a) Olfactory receptors in nasal passages are chiral proteins (2). (R)-carvone fits spearmint receptor binding site, triggering spearmint nerve signal (1). (S)-carvone fits caraway receptor site (1).	[4]
Q22	Q22: 22. Chirality of Substituted Biphenyls (Atropisomerism) 22. (a) Bulky ortho substituents (-NO ₂ and -CH ₃) create severe steric hindrance (2). Prevents free rotation around central C-C single bond, locking benzene rings perpendicular (1). Lacks plane of symmetry, forming stable chiral enantiomers (atropisomers) (1).	[4]
Q23	Q23: 23. Cahn-Ingold-Prelog (CIP) Priority Rules for R/S Assignment 23. (a) 1: -OH (O, atomic # 8) (1). 2: -CHO (C bonded to (O,O,H)) (1). 3: -CH ₂ OH (C bonded to (O,H,H)) (1). 4: -H (atomic # 1) (1). 23. (b) Priority 1 → 2 → 3 clockwise = R; counter-clockwise = S (1).	[5]
Q24	Q24: 24. Optical Activity of Cyanohydrin Hydrolysis 24. (a) Optically inactive / racemic mixture (hydrolysis does not alter configuration at chiral C2) (2). 24. (b) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{COOH} + \text{NH}_4\text{Cl}$ (2).	[4]

Q25	Q25: 25. Diastereomers vs Enantiomers Comparison 25. (a) Diastereomers are stereoisomers that are NOT mirror images of each other (2). Unlike enantiomers, diastereomers have DIFFERENT boiling points, melting points, solubilities, and NMR spectra (2).	[4]
Q26	Q26: 26. A* Challenge: Racemization Mechanism via Enolization 26. (a) Acid/base catalyzes reversible enolization: C2 proton is removed forming planar C=C enol intermediate (2). Re-protonation of planar enol C=C double bond occurs with equal 50:50 probability from top or bottom face (2). Converts pure (R)-isomer into 50:50 racemic mixture, losing optical activity (1).	[5]
Q27	Q27: 27. A* Challenge: Complete Multi-Step Synthesis of Pure Enantiomer 27. (a) Ethanal CH_3CHO + HCN (1). Oxynitrilase enzyme active site locks ethanal orientation, forcing CN^- to attack single face exclusively (1) $\rightarrow$ 100% (S)-cyanohydrin (1). 27. (b) $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ + $2\text{H}_2\text{O}$ + H^+ $\rightarrow$ $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ + NH_4^+ (2).	[5]
Q28	Q28: 28. A* Challenge: Chiral HPLC Separation using Cyclodextrin Stationary Phase 28. (a) Chiral stationary phase contains single enantiomer host molecules (beta-cyclodextrin cavities) (2). (+) and (-) enantiomers bind with different binding affinities (forming transient diastereomeric complexes) (2). Enantiomer with weaker binding elutes faster (shorter retention time t_R), achieving baseline resolution (1).	[5]
Q29	Q29: 29. A* Challenge: Stereochemistry of Electrophilic Addition to Alkenes 29. (a) Br_2 adds via cyclic bromonium ion intermediate (2). Subsequent attack by Br^- occurs strictly from opposite face (anti-addition) (1). For trans-but-2-ene, anti-addition forms meso-2,3-dibromobutane (internal symmetry) (1). For cis-but-2-ene, anti-addition forms (2R,3R) and (2S,3S) enantiomeric pair (racemate) (1).	[5]
Q30	Q30: 30. A* Challenge: Polarimetry Enantiomeric Excess (ee) Calculation 30. (a) % ee = $(+9.2 / +23.0) \times 100\%$ = 40.0% ee of (R)-isomer (2). 30. (b) 40% is pure (R). Remaining 60% is racemic (30% R + 30% S). Total % (R) = 40 + 30 = 70.0% (2). Total % (S) = 30.0% (1).	[5]
Q31	Q31: 31. A* Challenge: Stereochemistry of Epoxidation and Ring Opening 31. (a) Epoxide ring creates a rigid 3-membered ring on one face (2). Protonation forms oxonium ion. Water nucleophile must attack from the OPPOSITE face (backside $\text{S}_\text{N}2$ attack) to open ring (2). Leads exclusively to trans-1,2-diol stereochemistry (1).	[5]
Q32	Q32: 32. A* Challenge: Kinetic Resolution of Enantiomers 32. (a) Lipase active site matches (R)-ester, lowering its activation energy for hydrolysis (2). At 50% overall conversion, all (R)-ester is converted to (R)-carboxylic acid (2). Unreacted remaining ester is 100% pure (S)-ester (% ee > 99%) (1).	[5]
Q33	Q33: 33. A* Challenge: Optical Activity of Chiral Sulfoxides and Amines 33. (a) Sulfur atom is bonded to 3 different groups plus a non-bonding lone pair (4th group) in tetrahedral geometry (2). High inversion barrier (>100 kJ mol ⁻¹) prevents umbrella flip at 25 °C (1). 33. (b) Low inversion barrier (~25 kJ mol ⁻¹) allows nitrogen lone pair to flip rapidly through planar sp ² transition state at room temp, interconverting enantiomers instantly (2).	[5]
Q34	Q34: 34. A* Challenge: Chiral Pool Synthesis from Natural L-Amino Acids 34. (a) Chiral pool refers to readily available, inexpensive natural chiral building blocks (amino acids, sugars, terpenes) (2). 34. (b) Secondary amine of L-proline reacts with ketone forming a chiral enamine (1). Carboxylic acid group of proline hydrogen-bonds to incoming aldehyde, directing attack to a single face (2).	[5]
Q35	Q35: 35. A* Challenge: Stereochemical Proof of $\text{S}_\text{N}2$ Inversion using Radioactive Iodide 35. (a) $\text{S}_\text{N}2$ exchange: (+)-2-iodooctane + I^- $\rightarrow$ (-)-2-iodooctane (inversion) (2). ONE substitution event converts (+)-molecule into (-)-molecule (1). The (-)-molecule cancels ONE remaining (+)-molecule in polarimeter reading (1). Therefore, each individual substitution event destroys TWO units of optical rotation (rate of loss of rotation = 2 x rate of exchange) (1).	[5]

Q36	Q36: 36. A* Challenge: Optical Activity of Condensation Polyesters with Chiral Monomers 36. (a) Polymer chain contains 100% (S)-chiral centres in regular sequence, allowing tight helical chain packing and high crystallinity (3). 36. (b) PLLA has random R and S centres, preventing regular chain packing → amorphous, lower melting point, faster degradation (2).	[5]
Q37	Q37: 37. A* Challenge: Optical Activity of Halogenation of Chiral Alkanes 37. (a) 2,2-dichlorobutane is optically inactive (C2 has two Cl atoms, no chiral centre) (2). 37. (b) Cl. radical abstracts H from C2, forming trigonal planar .C(Cl)(CH ₃)CH ₂ CH ₃ radical (1). Cl ₂ attacks planar radical from top or bottom face with equal 50:50 probability (1), forming racemic 2,2-dichlorobutane (1).	[5]
Q38	Q38: 38. A* Challenge: Circular Dichroism (CD) Spectroscopy of Protein Secondary Structures 38. (a) Chiral peptide backbone chromophores absorb left- and right-circularly polarised UV light differently (2). Alpha-helix shows characteristic double negative minima at 208 nm and 222 nm (1). Beta-sheet shows negative minimum at 218 nm (1). Random coil shows negative minimum at 195 nm (1).	[5]
Q39	Q39: 39. A* Challenge: Optical Activity of Grignard Reactions with Chiral Aldehydes 39. (a) Substrate already possesses a chiral centre at C2 (1). Top and bottom faces of C=O group are sterically and electronically non-equivalent (diastereotopic faces) (2). Grignard nucleophile attacks the less hindered face preferentially (Felkin-Anh conformation) (1), yielding unequal amounts of diastereomeric alcohol products (1).	[5]
Q40	Q40: 40. A* Challenge: Optical Activity of Sugar Mutarotation (D-Glucose) 40. (a) Cyclic hemiacetal ring opens reversibly to form open-chain aldehyde intermediate (2). Rotation around C1-C2 bond and re-cyclisation forms beta-anomer (-OH equatorial) and alpha-anomer (-OH axial) (2). At equilibrium, 36% alpha and 64% beta exist, giving weighted average rotation $[\alpha] = (0.36 \times 112) + (0.64 \times 18.7) = +52.7^\circ$ (1).	[5]
Q41	Q41: 41. A* Challenge: Optical Activity of Hydroboration-Oxidation vs Oxymercuration of Alkenes 41. (a) BH ₃ adds across C=C double bond in a concerted 4-center transition state (1). Both B and H add to the SAME face of alkene (syn-addition) (2). 41. (b) Product is a racemic mixture of (1R,2R) and (1S,2S) enantiomers (optically inactive) because BH ₃ attacks top/bottom faces equally (2).	[5]
Q42	Q42: 42. A* Challenge: Stereochemistry of Beckmann Rearrangement of Chiral Oximes 42. (a) Protonation/activation of oxime -OH group → -OH ₂ ⁺ (1). Intramolecular 1,2-shift of R group to nitrogen occurs simultaneously with loss of H ₂ O (concerted mechanism) (2). Migrating R group never detaches as a free carbocation; bond to nitrogen forms on the same face as original bond to carbon, preserving stereochemistry 100% (2).	[5]
Q43	Q43: 43. A* Challenge: Chiral Auxiliary Directed Asymmetric Synthesis (Evans Oxazolidinones) 43. (a) 1. Attach chiral auxiliary to achiral reactant (1). 2. Perform reaction; auxiliary sterically blocks one face, directing reaction to form 99% single diastereomer (2). 3. Cleave chiral auxiliary under mild conditions (1). 4. Recover intact auxiliary for re-use, leaving pure single enantiomer product (1).	[5]
Q44	Q44: 44. A* Challenge: Optical Activity of Oxymercuration-Demercuration of Chiral Alkenes 44. (a) Hg ₂ ⁺ forms a 3-membered cyclic mercurinium ion (2). Prevents formation of free carbocation, suppressing 1,2-hydride shifts (1). 44. (b) Product is a 50:50 racemic mixture at C2 (optically inactive) (2).	[5]
Q45	Q45: 45. A* Challenge: Optical Activity of Ozonolysis of Chiral Terpenes 45. (a) Ozone undergoes 1,3-dipolar cycloaddition strictly across C=C double bonds (2). Does not break bonds attached to saturated chiral sp ³ carbon atoms (1). 45. (b) Product retains the original (R)-chiral centre and is optically active (2).	[5]
Q46	Q46: 46. A* Challenge: Optical Activity of Wittig Reaction with Chiral Ketones 46. (a) (R)-3-methylcyclohexanone + Ph ₃ P=CH ₂ → (R)-1-methylene-3-methylcyclohexane + Ph ₃ P=O (2). 46. (b) Wittig reaction replaces C=O with C=CH ₂ at C1 (1). Chiral centre at C3 is unaffected (1). Product maintains asymmetric carbon at C3 and remains optically active (1).	[5]

Q47	Q47: 47. A* Challenge: Stereochemistry of Sharpless Asymmetric Epoxidation 47. (a) Titanium forms a rigid dimeric complex with chiral diethyl tartrate DET ligands and allylic alcohol (2). (+)-DET forces oxygen delivery exclusively from top face of C=C double bond (>95% ee) (1.5). (-)-DET forces oxygen delivery exclusively from bottom face (>95% ee) (1.5).	[5]
Q48	Q48: 48. A* Challenge: Resolution of Racemic Base via Chiral Tartaric Acid 48. (a) (+)-amine.(+)-tartrate and (-)-amine.(+)-tartrate are DIASTEREOMERS (3). Diastereomers have different solubilities in ethanol. 48. (b) Filter off less soluble diastereomeric salt. Treat with strong base NaOH: (-)-amine.(+)-tartrate + NaOH → (-)-1-phenylethylamine (free base) + Na ₂ tartrate (2).	[5]
Q49	Q49: 49. A* Challenge: Optical Activity of Electrocyclic Reactions (Woodward-Hoffmann Rules) 49. (a) Conservation of orbital symmetry (Woodward-Hoffmann rules) (1). 4n pi electron thermal electrocyclic reactions require CONROTATORY ring opening (both methyl groups rotate in SAME direction) (2). Conrotatory opening of trans-3,4-dimethylcyclobutene yields single (2E,4E)-hexa-2,4-diene stereoisomer exclusively (2).	[5]
Q50	Q50: 50. A* Challenge: Complete Chirality & Stereochemistry Master Synthesis 50. (a) 2,4-DNPH positive ⇒ carbonyl (aldehyde or ketone) (1). Tollens' negative ⇒ ketone ⇒ X is butanone CH ₃ COCH ₂ CH ₃ (1). Reduction of butanone gives Y = butan-2-ol CH ₃ CH(OH)CH ₂ CH ₃ (1). C2 in butan-2-ol is chiral ⇒ Y is optically active (1). 50. (b) Lab NaBH ₄ reduction is non-enzymatic: H ⁻ attacks planar C=O from top/bottom faces equally → 50:50 racemate (optically inactive) (1). Yeast contains chiral alcohol dehydrogenase enzyme which directs H ⁻ attack to single face exclusively → pure enantiomer (optically active) (1).	[6]

SECTION D: EMPIRICAL EXAMINER FAQS & COMMON TRAPS (10 FAQS) — 15A**FAQ #1: Chiral Centre Definition****Category: Definition Trap****COMMON EXAMINER TRAP / MISTAKE:**

Defining a chiral centre as a carbon with 4 bonds instead of 4 DIFFERENT groups.

MODEL ANSWER / EXAMINER EXPECTATION:

A chiral centre MUST be a carbon atom bonded to FOUR DIFFERENT atoms or groups of atoms. If any two groups are identical (e.g. $-\text{CH}_3$ and $-\text{CH}_3$), it is achiral.

FAQ #2: Racemic Mixture Optical Activity**Category: Racemate Trap****COMMON EXAMINER TRAP / MISTAKE:**

Thinking a racemic mixture rotates light because it contains chiral molecules.

MODEL ANSWER / EXAMINER EXPECTATION:

A racemic mixture contains EQUAL (50:50) amounts of (+) and (-) enantiomers. The rotation caused by (+) molecules is EXACTLY CANCELLED by (-) molecules $\Rightarrow$ net rotation = 0° .

FAQ #3: SN1 vs SN2 Stereochemical Outcome**Category: Mechanism Trap****COMMON EXAMINER TRAP / MISTAKE:**

Confusing which substitution mechanism yields inversion vs racemization.

MODEL ANSWER / EXAMINER EXPECTATION:

SN1 proceeds via a PLANAR CARBOCATION intermediate $\rightarrow$ equal attack from top/bottom face $\rightarrow$ RACEMIC MIXTURE (optically inactive). SN2 proceeds via BACKSIDE ATTACK $\rightarrow$ WALDEN INVERSION (retains optical purity).

FAQ #4: Nucleophilic Addition to Carbonyls**Category: Carbonyl Planarity****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting to mention planarity of $\text{C}=\text{O}$ group when explaining racemization.

MODEL ANSWER / EXAMINER EXPECTATION:

Always state: 1. Carbonyl $\text{C}=\text{O}$ group is TRIGONAL PLANAR. 2. Nucleophile ($:\text{CN}^-$ or $:\text{H}^-$) attacks with EQUAL PROBABILITY from above or below the plane. 3. Produces a 50:50 racemic mixture.

FAQ #5: Physical Properties of Enantiomers**Category: Enantiomer Properties****COMMON EXAMINER TRAP / MISTAKE:**

Claiming enantiomers have different boiling points or solubilities.

MODEL ANSWER / EXAMINER EXPECTATION:

Enantiomers have IDENTICAL physical properties (boiling point, melting point, density, solubility in achiral solvents). They differ ONLY in their direction of rotation of plane-polarised light and interaction with chiral environments/receptors.

FAQ #6: Diastereomers vs Enantiomers**Category: Stereoisomer Classification****COMMON EXAMINER TRAP / MISTAKE:**

Confusing diastereomers with enantiomers.

MODEL ANSWER / EXAMINER EXPECTATION:

Enantiomers are non-superimposable MIRROR IMAGES. Diastereomers are stereoisomers that are NOT mirror images (e.g. cis/trans isomers or meso compounds). Diastereomers HAVE DIFFERENT physical properties.

FAQ #7: Meso Compounds

Category: Meso Inactivity

COMMON EXAMINER TRAP / MISTAKE:

Assuming a molecule with 2 chiral centres is always optically active.

MODEL ANSWER / EXAMINER EXPECTATION:

If a molecule with 2 chiral centres has an INTERNAL PLANE OF SYMMETRY (meso compound), the rotation of one half cancels the other half => OPTICALLY INACTIVE.

FAQ #8: Resolving Racemic Mixtures

Category: Resolution Technique

COMMON EXAMINER TRAP / MISTAKE:

Attempting to separate enantiomers by simple fractional distillation.

MODEL ANSWER / EXAMINER EXPECTATION:

Enantiomers have identical boiling points and CANNOT be separated by fractional distillation. They must be reacted with a single enantiomer resolving agent to form DIASTEREOMERS, which have different physical properties.

FAQ #9: Thalidomide In Vivo Racemization

Category: Pharmacology Trap

COMMON EXAMINER TRAP / MISTAKE:

Assuming administering pure (R)-thalidomide completely eliminates teratogenic risk.

MODEL ANSWER / EXAMINER EXPECTATION:

Thalidomide racemizes IN VIVO (in human blood pH), rapidly interconverting (R)-thalidomide into the teratogenic (S)-enantiomer regardless of initial purity.

FAQ #10: Plane-Polarised Light Source

Category: Polarimetry Basics

COMMON EXAMINER TRAP / MISTAKE:

Confusing plane-polarised light with unpolarised light.

MODEL ANSWER / EXAMINER EXPECTATION:

Normal light vibrates in all planes perpendicular to direction of propagation. Passing light through a Polaroid filter restricts vibrations to a SINGLE PLANE (plane-polarised light).